Lab 1

Introduction to Programming - Lab 1

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Answer the theory questions. Follow the instructions in the Practical section for the programming exercises.

Theory

Question 1: Chained comparisons

Explain the meaning of `0 <= score <= 10`. Compare it with `score >= 0 and score <= 10` and the incorrect test `score >= 0 or score <= 10`. Give a value that reveals the mistake in the last condition.

Question 2: Swapping scalar values

Explain why `a = b` followed by `b = a` does not generally swap two previously different values. Describe a correct swap using a temporary variable. Trace your assignments for `a = 5` and `b = 9`; do not use multiple assignment.

Practical

Submission: provide one complete Python 3 script for each assigned exercise, together with your test cases. Explain important decisions as code comments. Screenshots are not required.

Scope: use variables, scalar values, strings, arithmetic, conditions, and `for` or `while` loops. Built-in input/output, conversion, `range`, `len`, `abs`, `min`, `max`, and basic string operations are permitted. Do not define functions or classes, import libraries, or use lists, tuples, dictionaries or sets. Algorithmic reasoning is allowed in this Practical section.

Input and output: each numeric value is provided on its own line, in the order stated in the question. A word, command or string occupies one whole line. Print only the requested result, without input prompts. In compact examples, `/` marks a line break; the slash is not part of the actual input or output. Test at least one boundary case and one ordinary case in addition to the example. Multiple valid outputs are accepted only where explicitly stated.

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Question 3: An increasing purchase price

The first notebook costs k credits, the second costs 2*k, and notebook i costs i*k. A student wants w notebooks and already has b credits. Find how many additional credits are needed, or zero if the budget is sufficient. Use a loop to accumulate the purchase cost.

Input: Read k, b and w.

Output: Print the additional credits needed.

Bounds: 1 <= k, w <= 1000; 0 <= b <= 10^9.

Example: input `4 / 15 / 3`; output `9`.

Question 4: Balancing one-unit and two-unit coins

Pay exactly n units using coins worth one and two units. Choose non-negative coin counts whose difference in counts is as small as possible. Print counts, not the total number of coins. Either optimal pair is acceptable if a tie occurs.

Input: Read n.

Output: Print the one-unit count followed by the two-unit count.

Bounds: 1 <= n <= 10000.

Example: input `8`; output `2 3`.

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